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Influence of adaptability of Serious Games on learning outcomes and the application of knowledge and skills in professional training

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Abstract

This paper explores the use of Serious Games in professional training through e-learning, aiming to enhance the teaching-learning process by employing adaptive Serious Games. Despite their long-standing application in professional development, there is a need for more evidence-based approaches to evaluate their effectiveness. The study utilises the Design Science Research (DSR) methodology to develop the Framework for Adaptive Serious Games (F_4ASG), which adapts game elements to achieve optimal learning and engagement outcomes. Grounded in existing literature, expert input, and user experiences, F_4ASG focuses on learning outcomes and adaptive mechanisms as key features for Serious Game development and assessment. The research began with a systematic literature review and proceeded with iterative evaluations of the framework via focus groups, expert interviews, and an experimental study using a developed Serious Game. These evaluations informed subsequent refinements, culminating in the third and final version of F_4ASG. The findings offer new insights and practical applications for the effective use of Serious Games in professional training. The F_4ASG serves as a comprehensive guide for creating and assessing adaptive Serious Games that foster improved learning and engagement.

Keywords: Serious Games; Professional Training; Framework; Development; Evaluation; E-learning

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1. Introduction

Professional training is necessary to transmit the specifications of the activities to be performed and to align these activities to the organisational goals. It can be seen that the scope of professional training goes beyond this primary sphere and sometimes covers behavioural skills or even transversal knowledge and skills, such as written communication.

In some cases, it is found that professional training needs to at least match the knowledge of incoming professionals to the organisation's needs or even compensate for formal education issues.

Thus, the importance of professional training is increasingly recognised, as are organisations' challenges in training their employees in a competitive and dynamic environment.

In a general and simplified way, ATD (What Is E-Learning? n.d.) defines e-learning as "... a structured course or learning experience delivered electronically".

Thus, e-learning presents itself as one of the solutions for professional training, using ICT to deliver its activities to students/collaborators.

In this context, e-learning is being used as a more economical and flexible solution to offer training as the technologies advance. In addition, e-learning provides the opportunity for activities in different formats, including games, using different technologies.

Increasing the efficiency of professional training is a key issue in organisations today. Reducing the time to train and the time to perform work activities satisfactorily has been a goal pursued for many years and emphasises the participants' engagement in training activities (Pistono, Santos, & Baptista, 2022c).

These same authors cite that "Serious Games appear as interesting solutions for bringing together training characteristics and potentially engaging resources" (Pistono, Santos, & Baptista, 2022a, p. 754)

As Almeida & Simoes (2019) highlighted, the impact of industry 4.0 in education is a new paradigm where "learning models are adapted and customized according to real-time learner profiles.".

In this sense, Serious Games can offer a personalised approach to professional training and be useful to introduce or reinforce concepts and skills using ICT already available to companies' employees.

These games can be deployed either in training initiatives, increasing engagement and optimising the time to train, or in reinforcement programs or even supporting the performance of professionals during their daily activities, reducing time to perform.

Serious Games have been used in professional training for decades, and the amount of research regarding their efficiency in this context has also been increasing.

However, the results of training using Serious Games are still questionable, and, in many cases, the results are not adequately measured or not aligned with the learning objectives.

The biggest challenge in getting Serious Games accepted in education (Sousa et al., 2016) is measuring their actual effects on learning. This challenge also exists in professional training.

Another challenge of applying Serious Games is adaptability (Lopes & Bidarra, 2011) because the lack of it can result in a loss of learning efficiency or the impossibility of applying the game again for the same users.

Also, there are still few existing evidence-based approaches to assess the contribution of Serious Games to learning, as Mayer (2014) highlighted.

Considering this context, this investigation's primary motivation is to improve the teaching-learning process in professional training, using adaptive Serious Games that can relate their characteristics, or game elements, to the learning outcomes.

This investigation, therefore, aims to study the application of Serious Games in results-oriented professional training.

Following the Design Science Research - DSR methodology, this investigation proposes a framework for Serious Games that predicts their adaptation and alignment to the expected learning outcomes, keeping high levels of engagement of the players.

From initial exploratory research, the investigation objective was determined, which is to design and propose a framework concerning the adaptation of Serious Games elements to improve:

- Player's experience and engagement in the game.
- Learning outcomes.

- Knowledge transfer to work situations.
- Application of the skills practised in the game in real scenarios during professional activities.

In order to achieve the proposed objective, the following research questions were identified:

- RQ1: What is the influence of the game elements?
- RQ2: How should the game elements be adapted?
- RQ3: How to classify and organise the game elements to adapt and meet previously established goals?

After this initial exploratory investigation, the starting point was a systematic literature review, which answered these three questions and enabled the advancement of the investigation and the initial proposal of a Framework for Adaptive Serious Games – F_4ASG.

This initial proposal was based on the literature relating to all the findings of this review, emphasising addressing the existing frameworks of Serious Games.

Then, continuing the investigation, also in the design and development phase, focus groups and interviews with Serious Games experts were conducted and then triangulated with the data from the systematic literature review to generate a second version of the framework.

This second version of the framework was applied to develop a Serious Game called "*Encontrando Tempo*" and then evaluate this game. This activity gave rise to a descriptive evaluation of the framework and resulted in suggestions for application.

The final evaluation of the framework was an experimental one, where the Serious Game "*Encontrando Tempo*" was applied to a group of users. The results were analysed and compared to the framework prescriptions to verify if the proposed artefact solves the initially identified problem.

2. Methodology

Some methodologies could address the theme proposed for this investigation, such as case study, action research, or Design Science Research - DSR.

However, as pointed out by Dresch, Lacerda, & José Antonio Valle Antunes Júnior (2020), DSR is more appropriate because this methodology has the characteristic of investigating how things should be, while the other methodologies, as the previously cited, are used to investigate how things are or behave.

Moreover, this investigation considers the design and development of a framework (artefact) as a fundamental part of the research, reinforcing the DSR research methodology choice.

In order to conduct this investigation study and to answer the research questions, the sequence of processes of the model presented by Peffers, Tuunanen, Rothenberger, & Chatterjee (2007) for DSR was adopted.

This investigation has begun at the first entry point, identifying the problem: Verification of the efficiency of using Serious Games in learning in professional training through e-learning.

Following the DSR methodology, initial exploratory research was carried out in its first stage, problem identification and motivation, and then a systematic literature review was conducted.

During the initial exploratory research, the motivation was improving the teaching-learning process in professional training by using Serious Games efficiently.

While the initial exploratory research enabled the identification of the problem, the systematic literature review was the basis for this investigation.

In the systematic literature review, carried out with the methodological rigour indicated by Kitchenham (2004), three cycles of searches were performed, which after applying the inclusion, exclusion, and classification criteria, resulted in 53 final publications.

In this review, the review questions were the three research questions of this thesis, providing further elucidation on them and the treatment given to the related issues.

From the results of the systematic literature review, in the next stage of the DSR methodology, definition of the objectives of the solution, the objective initially proposed for the investigation was confirmed, and the objectives of the solution to be proposed were described in the form of the framework's purposes.

Next, the Framework for Adaptive Serious Games – F_4ASG was developed in the design and development stage.

This stage of the investigation comprised the following activities:

- First version of the Framework for Adaptive Serious Games – F_4ASG, based on the systematic literature review and on the qualitative analysis of the existing frameworks for Serious Games.
- Focus groups, conducted with groups of developers of training solutions that use game strategies and of professionals who use Serious Games in professional training, such as, for example, HR professionals, L&D, trainers, and training facilitators.
- Interviews with Serious Games experts with experience in developing and applying Serious Games in professional training or academic education.
- Second version of the Framework for Adaptive Serious Games – F_4ASG, from the triangulation of the data obtained with the different methodologies, that is, systematic literature review, focus groups and interviews.

Throughout this research, three evaluations were carried out, the first being at the design and development stage, as shown in Table 1.

Table 1. Evaluations of the Framework for Adaptive Serious Games – F_4ASG.

DSR Stage	Evaluation type	Design evaluation method	Evaluation description
Design and development	Artificial and formative	Analytical (static analysis) and testing (functional testing)	Triangulation among the results obtained in the Systematic Literature Review, the focus groups, and expert interviews.
Application	Artificial and formative	Descriptive (scenarios)	Applying the framework for developing and evaluating a Serious Game.
Evaluation	Naturalistic and summative	Experimental (controlled experiment)	Application of the Serious Game developed with the framework to users.

Table 1 also shows the design evaluation methods used among the forms listed by Hevner, March, Park, & Ram (2004, p. 86): Analytical, testing, descriptive and experimental.

The first evaluation is analytical since a static analysis was performed to triangulate the three methodologies to collect data (Hevner, March, Park, & Ram, 2004, p. 86). This evaluation is also classified as a functional testing type because during the triangulation, the intent was to identify confirmations, complements and divergences (Hevner, March, Park, & Ram, 2004, p. 86).

Thus, in the design and development stage, the first evaluation of the framework was carried out, as a formative and artificial one, based on the data collected in the focus groups and interviews. This evaluation resulted in a new framework version after triangulation of the data obtained from these methodologies and the systematic literature review.

As the following activity of the investigation, following the DSR, the Framework for Adaptive Serious Games – F_4ASG was applied to developing and evaluating a Serious Game.

The application of the framework for developing a Serious Game was described, following the three levels of detail provided by it, resulting in the game "*Encontrando Tempo*".

The framework was then used to evaluate this game and identify its potential concerning each dimension of the framework and thus enable improvements to the game.

In this way, the framework's application can be considered a descriptive evaluation and, like the previous evaluation, formative and artificial. This evaluation is of the descriptive type based on the construction of a scenario to demonstrate the framework's utility (Hevner, March, Park, & Ram, 2004, p. 86).

In the evaluation stage of the DSR, an experiment was proposed, based on the Serious Game developed in the previous stage of the investigation, as an experimental evaluation of the proposed framework.

Thus, the third planned evaluation is an experimental one, in the form of a controlled experiment (Hevner, March, Park, & Ram, 2004, p. 86). This evaluation resulted from the analysis of the players' interactions and results in the Serious Game developed from the framework application.

This evaluation, naturalistic and summative, aims to verify whether the proposed framework solves the problem initially identified in this investigation, that is, to verify the efficiency of using Serious Games in learning in professional training through e-learning.

As the Framework for Evaluation in Design Science Research - FEDS (Venable, Pries-Heje, & Baskerville, 2016) foresees, artefact evaluations were conducted throughout the investigation.

The evaluation strategy used in this investigation resembles the "*Quick and Simple*" strategy described by these authors.

The DSR communication stage corresponds to the activities carried out throughout the investigation, with the dissemination of intermediate results in conferences and journals, as listed in Table 2, and the investigation results in the thesis.

Table 2. Communication in journals and conferences.

Journal, conference or book	Contribution
21st Conference of the Portuguese Association for Information Systems (CAPSI 2021)	Influence of adaptability of Serious Games on learning outcomes and the application of knowledge and skills in professional training (Pistono, Santos, & Baptista, 2021b)
Journal of Digital Media & Interaction (JDMI)	A Review of Adaptable Serious Games Applied to Professional Training (Pistono, Santos, & Baptista, 2021a)
Gamers4Nature	Framework for development and evaluation of adaptable Serious Games in professional training (Pistono, Santos, & Baptista, 2022b)
International Conference on Industry Sciences and Computer Sciences Innovation 2022 (iSCSi 2022)	A qualitative analysis of frameworks for training through Serious Games (Pistono, Santos, & Baptista, 2022c)
1st LE@D Researchers Meeting - Innovation and Science (Meet.UP22)	Influência da adaptabilidade de jogos sérios nos resultados de aprendizagem e na aplicação de conhecimentos e habilidades na formação profissional (Pistono, Santos, & Baptista, 2022a)
International Conference on Technology and Innovation in Learning, Teaching and Education 2022 (TECH EDU 2022)	An Initial Framework for Adaptive Serious Games Based on a Systematic Literature Review (de Araujo Pistono, Santos, & Baptista, 2022)
18th Iberian Conference on Information Systems and Technologies (CISTI 2023)	Proposal of a framework for adaptive Serious Games using Design Science Research methodology (De Araujo Pistono, Santos, & Baptista, 2023)
Creating Learning Organizations Through Digital Transformation	Book chapter: "Improving Learning and Engagement in Professional Training Using Serious Games" (de Araujo Pistono, 2024a)
Computer Applications in Engineering Education	Framework for adaptive serious games (Pistono, dos Santos, Baptista, & Mamede, 2024b)

Throughout the thesis, three versions of the Framework for Adaptive Serious Games – F_4ASG were developed, as shown in Figure 1. In this figure, the activities conducted in the investigation can also be observed according to the stages of the DSR.

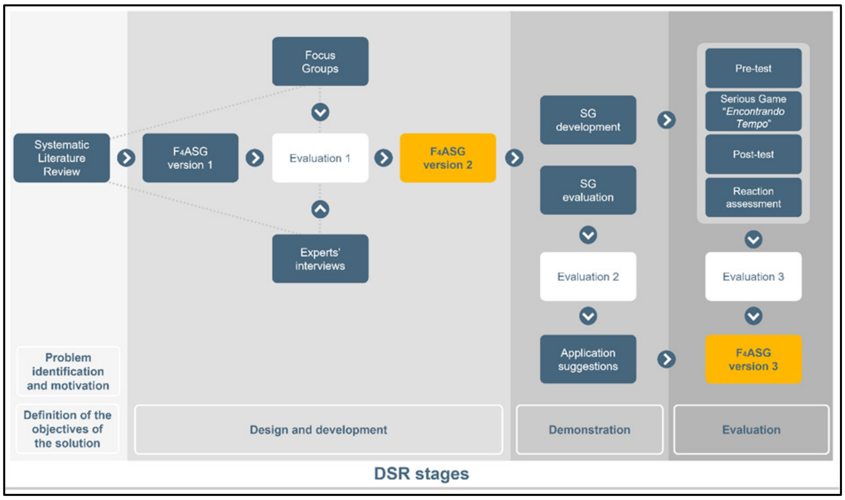

Fig. 1. Framework development

This figure highlights the evolution of the Framework for Adaptive Serious Games – F_4ASG:

- Version 1 resulted from the systematic literature review.
- Version 2 arose from data triangulation from a systematic literature review, focus groups, and expert interviews.
- Version 3 is the incorporation of the framework's application suggestions and the results of the experimental evaluation.

Figure 2 shows the final Framework for Adaptive Serious Games – F_4ASG.

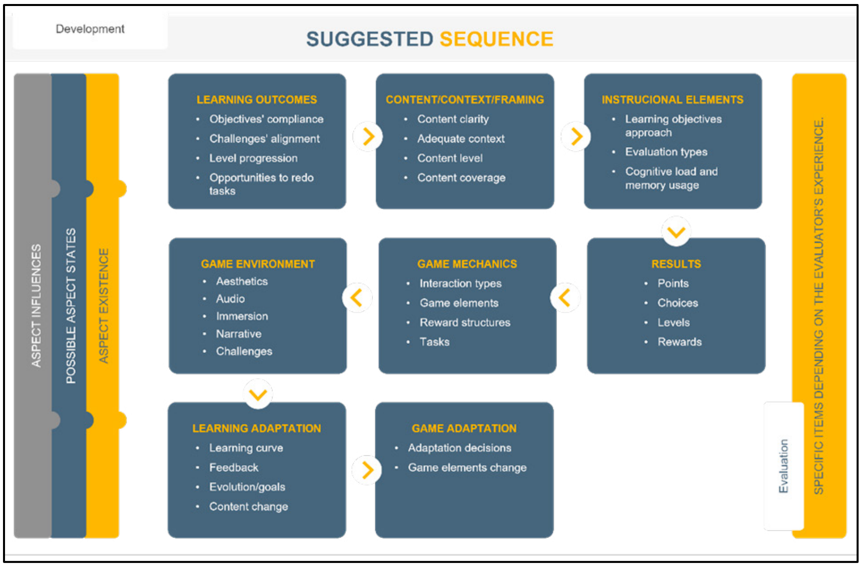

Fig. 2. Framework for Adaptive Serious Games – F_4ASG version 3

3. Conclusions

At the end of this research, it is important to present the main conclusions related to each research question:

RQ1: What is the influence of the game elements?

Serious Games are increasingly used for professional training to enhance participant engagement, which is crucial in achieving successful outcomes. However, further research is required to establish empirical evidence for the effectiveness of game elements in this context.

The balance between the different elements of a Serious Game is critical to its success, as it can determine the players' results.

Achieving this balance is a significant challenge, given that Serious Games must balance instruction and game elements.

It is essential to avoid distracting players with non-essential game elements or overwhelming their cognitive load with peripheral events.

Conversely, the presentation of content or tests should not disrupt the game's flow.

Therefore, it is crucial to achieving a balance among all the Serious Game elements to ensure engagement and successful learning outcomes, and, ideally, to keep the players in a state of flow.

RQ2: How should the game elements be adapted?

The adaptation forms of a Serious Game must predict the learning and the game dimensions, so it can maintain the elements' balance and promote the best choices to improve the game outcomes.

Potentially, all game elements could be adaptable, using parametrisation or changes during the game.

Adaptation can have two main focuses in a Serious Game: Learning and Game.

The first, concerning achieving better learning outcomes, must consider the possibilities to adapt the learning curve, feedback content, the goals that guide the player to evolve in the game and even the content presented.

The game adaptation concerning the game elements and their states can be represented in adaptation decisions, and the game elements change themselves.

The most common adaptations in games using artificial intelligence are Dynamic Difficulty Adjustment – DDA and the adjustment of contents and mechanics.

So, it is possible to adapt the Serious Games elements to achieve desired learning outcomes and to improve participant engagement.

It is noteworthy that, if possible, rebalance the Serious Game elements after application to at least one pilot group of players so that the adaptive characteristics and the individual game elements are adjusted.

RQ3: How to classify and organise the game elements to adapt and meet previously established goals?

This investigation has shown that it is possible to classify and organise Serious Games' elements to adapt their states and foster learning.

Developing a Serious Game based on the best practices of instruction and game theories is possible.

The most important is to focus on the outcomes and predict the necessary adaptation or rebalancing of each Serious Game element.

This challenge is increasingly present because participants of training activities have less and less time to train and more urgency to perform new work activities. At the same time, materials with increasingly attractive formats and a direct communication approach are present in their daily lives.

Although it is a challenging task, adaptive Serious Games have the potential to promote engagement and learning opportunities to grant better learning outcomes.

Moreover, the problem initially identified, i.e. "*Verification of the efficiency of the use of Serious Games in learning in professional training through e-learning*", was solved by verifying the possibility of addressing professional training using an adaptive Serious Game developed using the F_4ASG, as well as verifying its efficiency through evaluation also using this framework.

In this context, the Framework for Adaptive Serious Games – F_4ASG can be considered complete and effective, enabling us to meet the initial requirements and constraints.

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